

★ Unit - 2 : Communication Methodologies

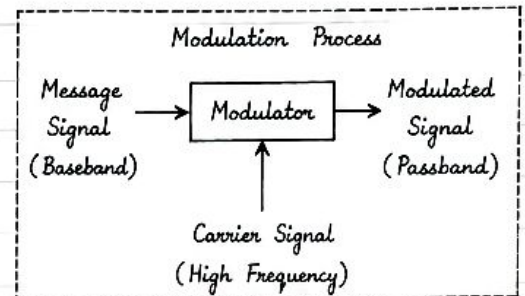
Need for Modulation in Communication System

Modulation is the process of varying one or more parameters of a high frequency carrier signal in accordance with the information (baseband) signal. It is an essential technique in communication systems. Without modulation, the communication would be ineffective and inefficient.

1. What is Modulation ?

Modulation is the process of impressing the information signal (baseband) onto a high frequency carrier signal by varying its amplitude, frequency or phase.

The modulated signal is then transmitted through the communication channel.

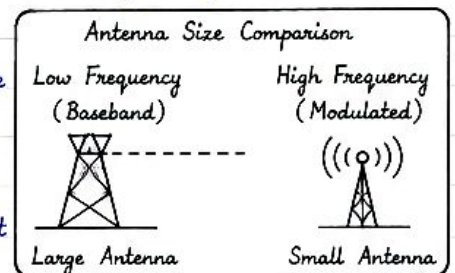


2. Need for Modulation

Modulation is necessary in a communication system due to the following reasons :

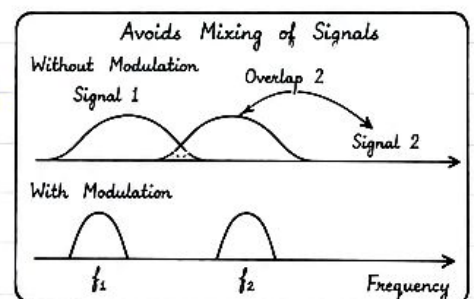
① Efficient Radiation of Energy :

For efficient radiation, the antenna length should be comparable to the wavelength of the signal. Since baseband signals have very low frequency (long wavelength), they require very large antennas. Modulation shifts the signal to high frequency (short wavelength) which allows the use of practical antenna size.



② Avoids Mixing of Signals :

If baseband signals are transmitted, different signals may overlap and mix up. Modulation shifts each signal to a different carrier frequency, so multiple signals can be transmitted simultaneously without interference.

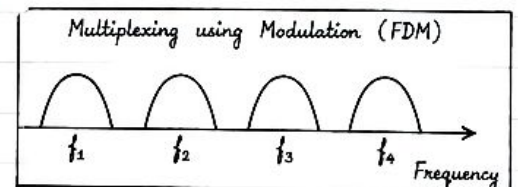


③ Increases Range of Communication :

High frequency signals (modulated) can travel longer distances as they are less affected by the size of antenna and can be effectively radiated.

④ Enables Multiplexing :

Modulation allows the use of techniques like Frequency Division Multiplexing (FDM), enabling multiple signals to be sent over the same channel by using different carrier frequencies.



⑤ Reduces Effect of Noise and Interference :

High frequency modulated signals are less affected by low frequency noise and interference.

⑥ Compatible with Bandpass Channel :

Most communication channels (like radio, microwave, satellite) are bandpass in nature and can transmit only a range of frequencies. Modulation shifts the baseband signal to this usable range.

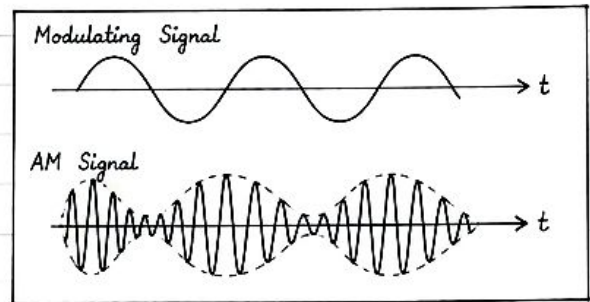
Summary

Modulation is essential for practical and efficient communication. It helps in efficient radiation, avoids signal mixing, increases range, enables multiplexing, reduces noise effect and is compatible with bandpass channels.

• Concepts : AM, FM, PM, FSK, TSK, PCM

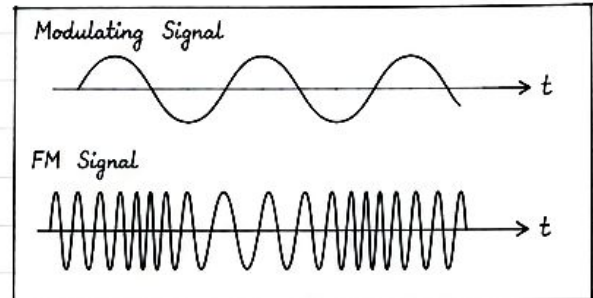
1. Amplitude Modulation (AM) :

In Amplitude Modulation, the amplitude of the carrier wave is varied in proportion to the instantaneous amplitude of the modulating signal, while the frequency and phase remain constant. It is simple and widely used in broadcasting (AM radio).



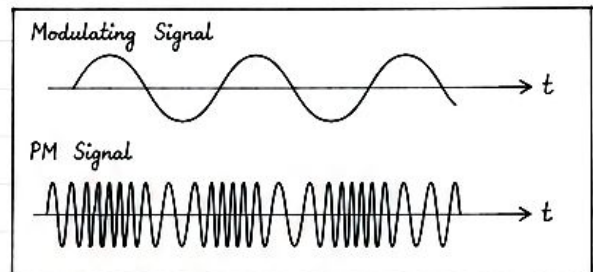
2. Frequency Modulation (FM) :

In Frequency Modulation, the frequency of the carrier wave is varied in proportion to the instantaneous amplitude of the modulating signal, while the amplitude remains constant. FM is less affected by noise and provides better sound quality. It is used in FM radio broadcasting.



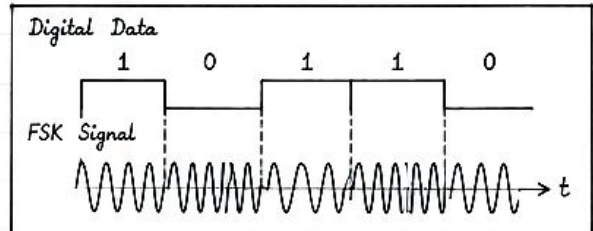
3. Phase Modulation (PM) :

In Phase Modulation, the phase of the carrier wave is varied in proportion to the instantaneous amplitude of the modulating signal, while the amplitude remains constant. PM is similar to FM and is used in some digital communication systems.



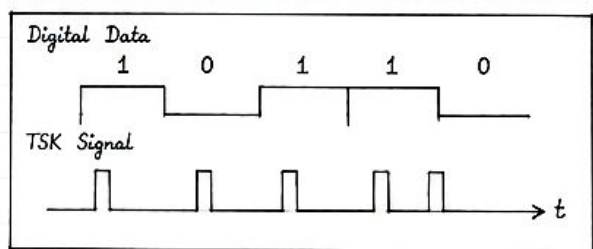
4. Frequency Shift Keying (FSK) :

FSK is a form of digital modulation in which digital data is transmitted by switching between two or more discrete frequencies. Each frequency represents a binary symbol (0 or 1). FSK is robust and used in wireless communication.



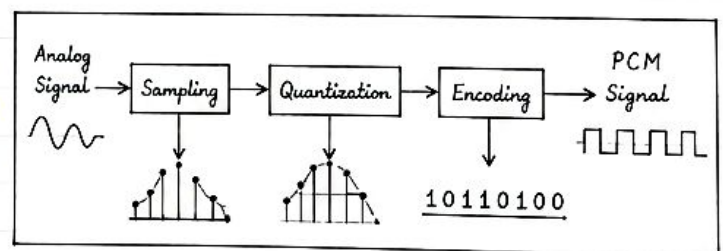
5. Time Shift Keying (TSK) :

TSK (also called Time Shift Keying or Time Position Modulation) is a digital modulation technique in which digital data is transmitted by shifting the position of a signal pulse in time. It is less affected by noise and easy to implement.



6. Pulse Code Modulation (PCM) :

PCM is a digital modulation technique in which the analog signal is first sampled, then quantised, and finally encoded into binary form. The binary data is transmitted as a digital signal. PCM is used in telephone systems, audio CDs, and digital communication.



★ Summary Table

Technique	What is Varied	What Remains Constant	Type	Main Use
AM	Amplitude	Frequency, Phase	Analog	AM Radio Broadcasting
FM	Frequency	Amplitude	Analog	FM Radio, TV Sound
PM	Phase	Amplitude	Analog	Digital Communication
FSK	Frequency (Discrete)	Amplitude	Digital	Wireless Communication
TSK	Time Position	Amplitude	Digital	Digital Data Transmission
PCM	Amplitude (via Sampling)	-	Digital	Telephony, Digital Systems

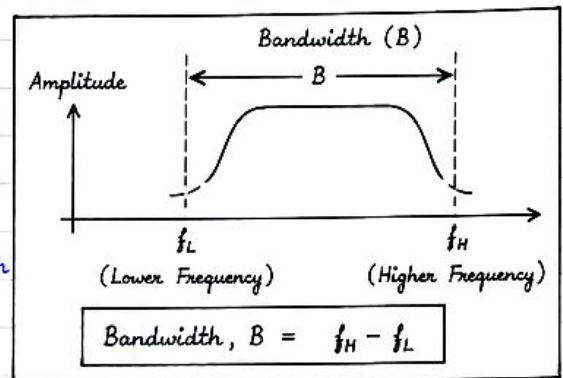
• Concept of Bandwidth and Channel Capacity of different Communication Systems such as Radio, Microwave etc.

1. Bandwidth :

Bandwidth is the range of frequencies that a communication channel can effectively pass.

It is the difference between the highest and lowest frequency components of the signal.

It is measured in Hertz (Hz). A wider bandwidth allows more information (higher data rate) to be transmitted.



2. Channel Capacity :

Channel capacity is the maximum rate (in bps) at which data can be transmitted over a communication channel with an arbitrarily small probability of error.

It depends on the bandwidth of the channel and the Signal to Noise Ratio (SNR) of the channel.

Factors Affecting Channel Capacity :

- Bandwidth (B)
- Signal to Noise Ratio (SNR)

Shannon's Theorem (Concept)

The channel capacity (C) increases with increase in bandwidth and SNR.

More bandwidth or higher SNR $\rightarrow$ Higher capacity.

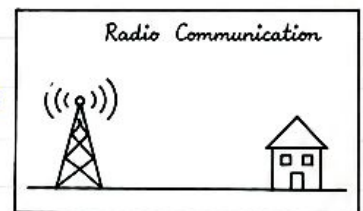
3. Channel Capacity in Different Communication Systems :

(i) Radio Communication :

Radio waves have low to moderate frequency (kHz to MHz range).

They have larger wavelength, can travel long distances and can penetrate buildings and obstacles.

- Bandwidth : Generally narrow to medium.
- Channel Capacity : Moderate (suitable for voice, limited data).

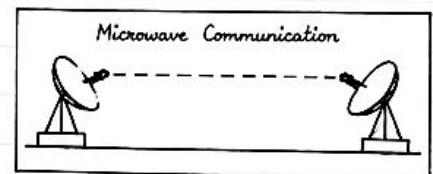


(ii) Microwave Communication :

Microwaves have higher frequency (1 GHz to 300 GHz).

They require line of sight (LOS) path.

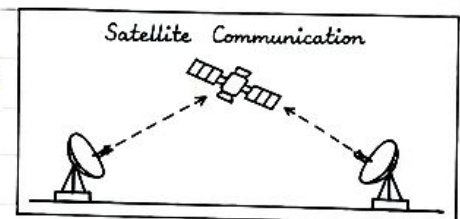
- Bandwidth : Wide.
- Channel Capacity : High (suitable for high speed data, TV, satellite links).



(iii) Satellite Communication :

Uses microwaves or radio waves for long distance communication through satellites.

- Bandwidth : Very wide (especially in modern satellites).
- Channel Capacity : Very high (used for TV broadcasting, internet, telephony).



4. Comparison Table :

System	Frequency Range	Bandwidth	Channel Capacity	Features / Use
Radio	kHz - MHz	Narrow to Medium	Moderate	Long range, can penetrate obstacles, used in AM, FM
Microwave	1 GHz - 300 GHz	Wide	High	Line of sight, used in TV, mobile, wireless links
Satellite	GHz Range	Very Wide	Very High	Long distance, used in TV broadcasting, internet

5. Summary :

Bandwidth determines how much information a channel can carry at a time, while channel capacity determines the maximum rate of reliable data transmission. Higher bandwidth and higher SNR result in higher channel capacity.

Key Point

Wider Bandwidth
+ Higher SNR
= Higher Channel Capacity

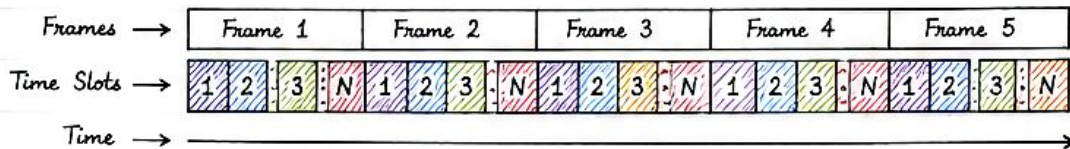
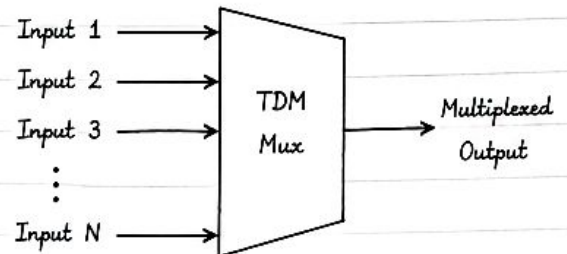
• Multiplexing Techniques

Multiplexing is a technique used to combine multiple signals or data streams so that they can be transmitted over a single communication channel. It improves the efficiency of the channel and increases its capacity.

The main multiplexing techniques are:

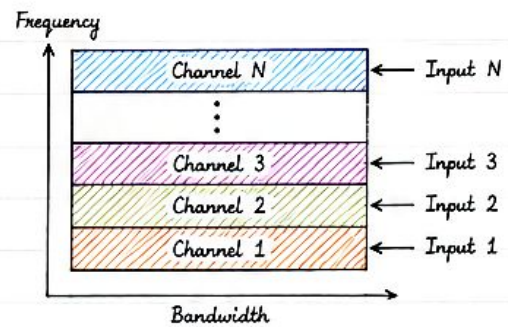
1. Time Division Multiplexing (TDM):

In TDM, the entire bandwidth of the channel is divided into equal time slots. Each input signal is assigned a specific time slot for transmission. The signals are transmitted one after another in a cyclic manner. It is commonly used in digital communication systems.



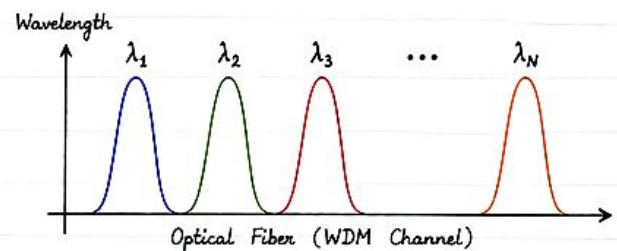
2. Frequency Division Multiplexing (FDM):

In FDM, the available bandwidth of the channel is divided into several non-overlapping frequency bands. Each input signal modulates a different carrier frequency. All the modulated signals are transmitted simultaneously over the same channel. It is used in analog communication systems.



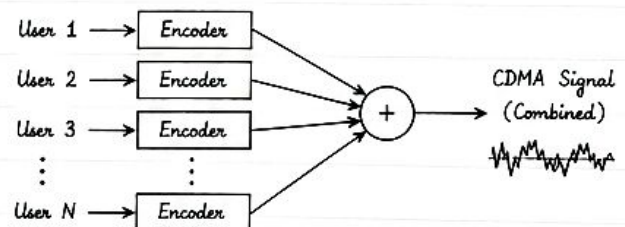
3. Wavelength Division Multiplexing (WDM):

WDM is similar to FDM, but instead of radio frequencies, it uses different wavelengths of light to transmit multiple signals through a single optical fiber. Each wavelength acts as a separate channel. It is widely used in fiber optic communication systems.



4. Code Division Multiple Access (CDMA):

In CDMA, all users transmit their data simultaneously over the same frequency band, but each user is assigned a unique code. The receiver uses the code to extract the intended signal and reject others. It is used in wireless communication systems like mobile networks.



Comparison Table

Technique	Basis	Transmission	Bandwidth Usage	Application	Type
TDM	Time	One at a time (in slots)	Full bandwidth in time slots	Digital systems	Digital
FDM	Frequency	All at same time	Divided into frequency bands	Radio, TV, Analog systems	Analog
WDM	Wavelength	All at same time	Divided into wavelengths	Fiber optic systems	Optical
CDMA	Code	All at same time	Same frequency band	Wireless, Mobile systems	Digital